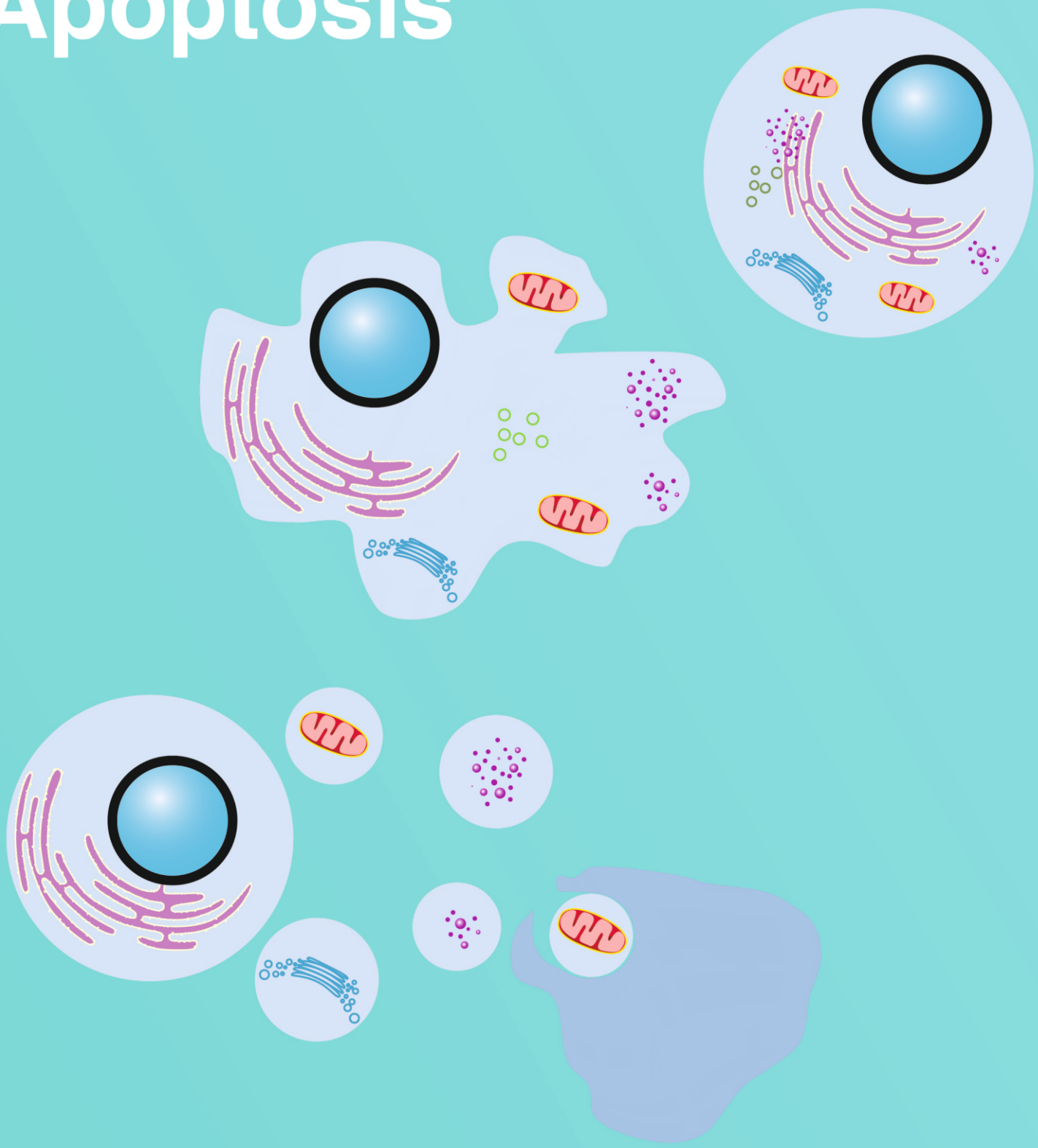
Solarbio®
LIFE SCIENCES

Sri Saradi Scientific

Apoptosis



Beijing Solarbio Science & Technology Co.,Ltd.

Introduction

Apoptosis is one specific form of programmed cell death (PCD),which is a form of cell death responsible for development and homeostasis of all multicellular organisms,Apoptosis is a normal natural or physiological process under regulation by a series of endogenous genes.Cells undergoing apoptosis usually exhibit a characteristics morphology, including nucleus concentration, cytoplasmic density increase,cytoplasmic condensation, nuclear membrane rupture,DNA fragmentation,cell shrinkage,membrane blebbing and formation of apoptosomes which are eventually phagocyto-sed by macrophages and will not trigger local inflammatory reaction. In terms of molecular mechanism, apoptotic protease cascade reaction initiated after caspase activation which leads to irreversible cell apoptosis.

Detection Technology

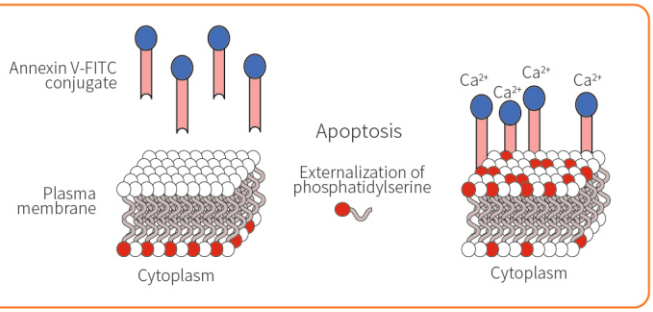
1. Location Analysis of TFAR 19
2. Annexin V Detection
3. Intracellular Glutathione Redox State（GSH detection）
4. Mitochondrial Transmembrane Potential Detection
5. Location Detection of Cytochrome C
6. Caspase Detection
7. DNA Fragmentation Detection
8. Fluorescent Dye Staining

Location Analysis of TFAR 19

TFAR19 (PDCD5) is a pro-apoptotic protein. The high expression and nuclear translocation of TFAR19 is one kind of new technique used for research the early apoptosis. The expression level and localization of TFAR19 protein were detected by immunoassay with **PDCD5 antibody(K005151P)**.

Annexin V Detection

Annexin V is a 35.8 KD Ca^{2+}-dependent phospholipid binding protein with high affinity for phosphatidylserine (PS), which is redistributed from the inner to the outer plasma membrane leaflet in apoptotic or dead cells.Annexin-V conjugated to fluorochromes (FITC, PE, etc.) as a fluorescent probe combined with nucleic acid dyes (PI, 7-AAD) is a double staining solution used for detecting apoptosis by flow cytometry or fluorescence microscopy. Early and late apoptotic cells and dead cells can be identified.



Catalog	Product name
CA1020	Annexin V FITC/PI Apoptosis Detection Kit
CA1030	Annexin V PE/7-AAD Apoptosis Detection Kit
CA1040	Annexin V Alexa Fluor488/PI Apoptosis Detection Kit
CA1050	Annexin V Alexa Fluor647/PI Apoptosis Detection Kit

Intracellular Glutathione Redox State（GSH detection）

There is ATP-dependent GSH transfer system in cell membrane which can be activated by apoptotic signal. GSH oxidation is the main elements of oxidant-mediated apoptosis, which leads to the decrease of mitochondrial membrane potential in early apoptotic, transferring cytochrome C from mitochondria to cytosol, and triggering caspase cascade reaction of apoptotic.

Catalog	Product name
BC1195	Micro Glutathione Peroxidase(GPX)Assay Kit
BC1165	Micro Glutathion Reductases(GR)Assay Kit
BC0355	Micro Glutathione S-transferase(GST)Assay Kit

Catalog	Product name
BC1185	Micro Oxidized Glutathione(GSSG)Assay Kit
BC1175	Micro Reduced Glutathione(GSH)Assay Kit

Mitochondrial Transmembrane Potential Detection

The decrease of mitochondrial membrane potential is a marker in the early apoptosis. Mitochondrial membrane potential is exsited in normal cells,some cationic(positively-charged) fluorescent dyes (such as JC-10) that accumulate in the mitochondrial matrix to form polymers which emit intense orange fluorescence (Em=570 nm). While cell apoptosis occurs, mitochondrial transmembrane potential depolarizes, dyes are released from mitochondria, the concentration of dyes decrease, and they exist in the form of monomers and emit green fluorescence (Em=520 nm). Therefore, the change of color can directly reflect the change of mitochondrial membrane potential. There are also other membrane potential detection probes: JC-1, Rhodamine 123 (Rh123), tetramethyl rhodamine ethyl ester (TMRE), etc.

Normal Cells	Apoptotic/Necrotic cells
ΔΨM	↓ΔΨM
JC-10 Aggregates	JC-10 Monomer
Em=570nm	Em=520nm
Orange	Green
FACS Channel=FL2	FACS Channel=FL1

Catalog	Product name
J8030	JC-1 Mitochondrial Membrane Potential Fluorescence Probe
M8650	Mitochondrial Membrane Potential Assay Kit(JC-1)
J8050	JC-10 Mitochondrial Membrane Potential Fluorescence Probe
CA1310	Mitochondrial Membrane Potential Assay Kit(JC-10)
R8030	Rhodamine 123
T7910	TMRE (Tetramethylrhodamine ethyl ester)

Location Detection of Cytochrome C

Cytochrome C is a signal molecule that plays an important role in apoptosis. Normally, it exists in the cave between the inner and outer membrane of mitochondria. Apoptotic signals stimulate it to release from mitochondria to cytosol, and then trigger caspase cascade reaction after combining Apaf-1. COX4 is a membrane protein located on the inner membrane of mitochondria, which still remains in the mitochondria when apoptosis occurs. Mitochondria can be separated from apoptotic and non-apoptotic cells, and then the location of cytochrome C and COX4 can be detected by Western blot using **cytochrome C antibody(K003448P) and COX4 antibody(K004618P)** respectively, to determine the occurrence of apoptosis.

Caspase Detection

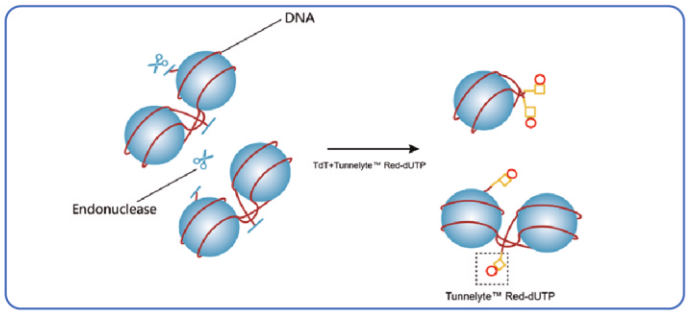
Caspase is a family of cysteine proteases that play essential roles in apoptosis. Caspase can self-activate and activate each other. Once the apoptotic process is triggered, cascade amplification effect appears. At present, there are 14 members in this family, which can be divided into three categories according to the function and location of upstream and downstream in cascade reactions: the first group is upstream apoptotic initiators, including caspase-2/8/9/10, etc. The second group is downstream apoptotic effector, including caspase-3/6/7. The third group is caspase-1/4/5/13/14, mainly involved in cytokine-mediated inflammatory response. It also plays an auxiliary role in the apoptotic pathway mediated by death receptors.

Catalog	Product name
BC3810	Caspase-1 Activity Assay Kit
BC3820	Caspase-2 Activity Assay Kit
BC3830	Caspase-3 Activity Assay Kit
BC3840	Caspase-4 Activity Assay Kit

Catalog	Product name
BC3850	Caspase-5 Activity Assay Kit
BC3860	Caspase-6 Activity Assay Kit
BC3880	Caspase-8 Activity Assay Kit
BC3890	Caspase-9 Activity Assay Kit

DNA Fragmentation Detection

Condensed chromatin can be fragmented in late apoptosis. Activation of CAD(caspase-activated DNase) leads to specific cleavage of the DNA at the internucleosomal linker sites between the nucleosomes, generating fragments of 180~200 base pairs known as DNA ladders. An alternative method for detecting DNA fragmentation involves the identification of nicks (or strand breaks) using the TdT dUTP nick end labeling (TUNEL) assay. The TUNEL technique is based on the ability of terminal deoxynucleotidyl transferase (TdT) to end-label nicks in the DNA brought about by caspase-activated deoxyribonuclease. TdT has the ability to label 3' blunt ends of double stranded DNA with deoxyuridine, independently of a template.



Catalog	Product name
T2190	TUNEL Apoptosis Assay Kit

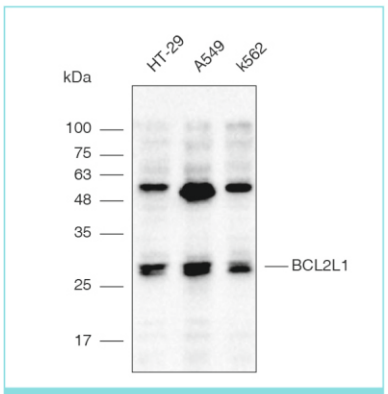
Fluorescent Dyes Staining

Catalog	Product name
CA1142	AO Fluorescent Staining Kit
CA1143	AO Staining Solution
CA1120	Hoechst 33342/PI Double Staining Kit
G3680	Hoechst Staining Kit
C0031	Hoechst 33342 Staining Solution

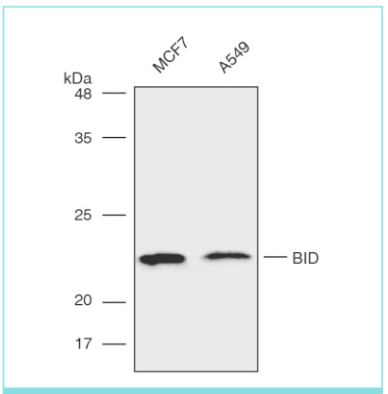
Catalog	Product name
B8040	Hoechst 33342 Fluorescent Staining Solution
C0020	Hoechst 33258 Staining Solution(ready-to-use)
B8030	Hoechst 33258 Fluorescent Staining Solution
C0065	DAPI
C0080	Propidium Iodide

Antibody

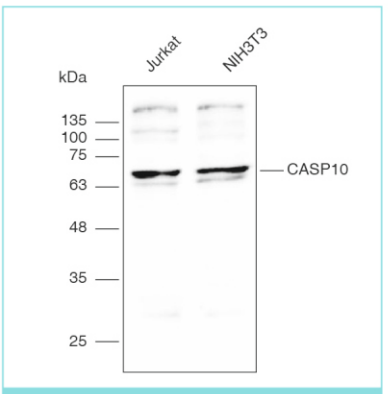
FasLG	CASP12	CTSS	HtrA2/Omi	MCL1	Phospho-p53-S46
AIF	CASP2	CTSV	IKBKB	MEK1	Phospho-p53-S9
AIFM1	CASP3	CTSZ	IKBKG	MLKL	PIDD
AIFM3	PolUHRF2	CYCS	IKKα	NFκB p65	PIK3CA
AKT1	CASP3	Cytochrome C	IKKβ	NFκB1	PIK3CB
AKT2	CASP3 PTAT	DDIT3	IL3	NFκB p105/p50	PIK3R1
AKT3	CASP7	DFFA	IL3RA	NGF	RAF1
APAF1	CASP8	DFFB	IκB-α(N-term)	NTRK1	RIPK3
ATF4	CASP9	DIABLO	JNK1	p38 MAPK	ROCK1
ATM	CASP6	DR5	JUN	p53	Smac
BAD	Cathepsin D	eIF2α	Lamin A+C	p85A	TAK1
BAK1	CFLAR	EndoG	Lamin A	PAK1	TNF
BAX	c-Fos	ERK1/2	Lamin B1	PAK2	TNFRSF10A
BCL2	cIAP2	FADD	Lamin B2	PAK4	TNFSF10
BCL2A1	c-Jun	FAS(C-term)	Lamin B	PARP	TP53AIP1
BCL2L1	Cleaved PARP	FASN	MAP2K2	PARP1	TRADD
BCL2L11	CSF2RB	GADD45	MAP2K7	PARP10	TRAF1
BCLXL	CTSB	GADD45α	MAPK1	PARP3	TRAF2
BID	CTSC	GAS2	MAPK10	PARP4	TRAIL
BIRC2	CTSD	GZMB	MAPK14	Phospho-FAS-Y291	XIAP
BIRC5	CTSK	HRAS	MAPK8	Phospho-MAP3K5-S1033	
CASP10	CTSL	HTRA2	MAPK9	Phospho-NFκB p105-S927	



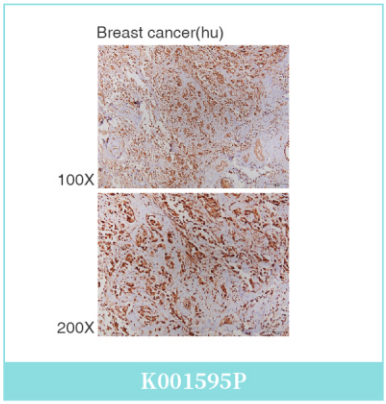
K001595P



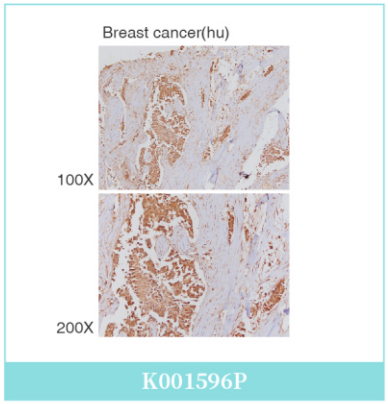
K001596P



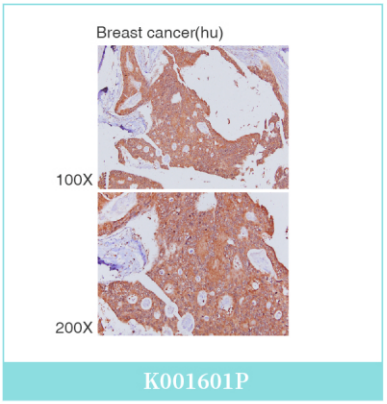
K001601P



K001595P



K001596P



K001601P

Small Molecular Compound

Catalog	Product name
IZ0050	Z-VAD-FMK
IZ0040	Z-IETD-FMK
IG0430	Gossypol (acetic acid)
IG0050	Gambogic Acid
IG0420	Gossypol
IC0570	Costunolide
IA0130	Alantolactone
IA0170	Aloin
IA0620	Atractylenolide III
IE0110	Epibrassinolide
IG0400	Glycitein
II0160	Isoalantolactone
IS0030	Sanguinarine
IS0040	Sanguinarine (chloride)
IB0190	Betulinic acid

Catalog	Product name
II0170	Isobavachalcone
IG0280	Ginsenoside Rh2
IB0030	Baohuoside I
IC0440	Cisplatin
IH0150	Hydroxyurea
IG0110	Genistein
IK0020	Kaempferol
IS0190	Sodium Butyrate
ID0380	Doxorubicin (hydrochloride)
II0140	Irinotecan
II0150	Irinotecan (hydrochloride trihydrate)
IE0270	Etoposide
IB0120	Berberine (chloride)
C6700	CCCP
IB0080	Benzoylpaeoniflorin

Cell Function Assay

Catalog	Product name
CA1510	DNA Content Quantitation Assay Kit(Cell Cycle)
CA1630	Calcein-AM/PI Double Staining Kit
M1020	MTT Cell Proliferation and Cytotoxicity Assay Kit
CA1210	CCK-8 Cell Proliferation and Cytotoxicity Assay Kit
S1076	CFSE(CFDA-SE)

Catalog	Product name
G0170	Autophagy Staining Assay Kit(MDC)
G1310	Neutral Red solution(Specific for living cells)
D6470	DCFH-DA(ROS Fluorescent Probe)
F8840	Fluo 3-AM
F8500	Fluo 4-AM

Representative Papers

Journal	IF	Product	PMID
Advanced Materials	30.8	Hoechst 33342 Stain solution(ready-to-use)(C0030)	35315546
Molecular Plant	21.949	Reduced Glutathione(GSH) Content Assay Kit/Oxidized Glutathione(GSSG) Content Assay Kit(BC1175/BC1185)	35799449
Journal of the American Chemical Society	16.383	Mitochondrial Membrane Potential Kit(JC-10 Assay)(CA1310)	35357824
ACS Nano	15.881	TUNEL Apoptosis Detection Kit(T2190)	35025206
ACS Nano	15.881	Hoechst 33342/PI Double Stain Kit(CA1120)	34978417
Angewandte Chemie International Edition	15.336	Reduced Glutathione(GSH) Content Assay Kit(BC1175)	35238124
Clinical Cancer Research	12.53	TUNEL Apoptosis Detection Kit/Reduced Glutathione(GSH) Content Assay Kit(T2190/BC1175)	35227287

Journal	IF	Product	PMID
Nanoscale Horizons	10.989	Annexin V-FITC Apoptosis Detection Kit(CA1020)	35048915
Materials Today Bio	10.761	Reduced Glutathione(GSH) Content Assay Kit(BC1175)	35601897
Journal of Hazardous Materials	10.588	Oxidized Glutathione(GSSG) Content Assay Kit(BC1180)	35427968
Journal of Nanobiotechnology	10.435	CCK-8 Cell Proliferation and Cytotoxicity Assay Kit/Reactive Oxygen Species Assay Kit(CA1210/CA1410)	35073918
ACS Applied Materials & Interfaces	10.383	Reduced Glutathione(GSH) Content Assay Kit(BC1175)	35735117
ACS Applied Materials & Interfaces	10.323	Annexin V-FITC Apoptosis Detection Kit(CA1020)	35511491
Science of the Total Environment	7.963	Fluo-3, AM Calcium Ion Detection Kit/Caspase-3 Activity Assay Kit(CA1180/BC3830)	35051481
Biomedicine & Pharmacotherapy	6.529	Reactive Oxygen Species Assay Kit(CA1410)	35290888
Oxidative Medicine and Cellular Longevity	6.5	Annexin V-FITC Apoptosis Detection Kit(CA1020)	35368874
Oxidative Medicine and Cellular Longevity	6.5	Mitochondrial Membrane Potential Assay Kit with JC-1/Reactive Oxygen Species Assay Kit(M8650/CA1410)	35592531

Apoptosis Pathway

Cell apoptosis, sometimes called programmed cell death, is a cellular self-destruction method to remove old and damaged cells to maintain physiological homeostasis. Apoptosis also occurs as a defense mechanism such as in immune reactions or when cells are damaged by disease or noxious agents. Apoptosis is controlled by many genes and involves three fundamental pathways: the extrinsic pathway (death receptor pathway), the intrinsic pathway (mitochondrial pathway) and the endoplasmic reticulum pathway. Each pathway interacts with each other to regulate the apoptotic process.

Extrinsic Pathway (death receptor pathway)

1. Fas/FasL and TRAILR/TRAIL pathway
2. TNFR1/TNF pathway

Intrinsic Pathway (mitochondrial pathway)

Endoplasmic Reticulum Pathway

Bcl-2 Superfamily

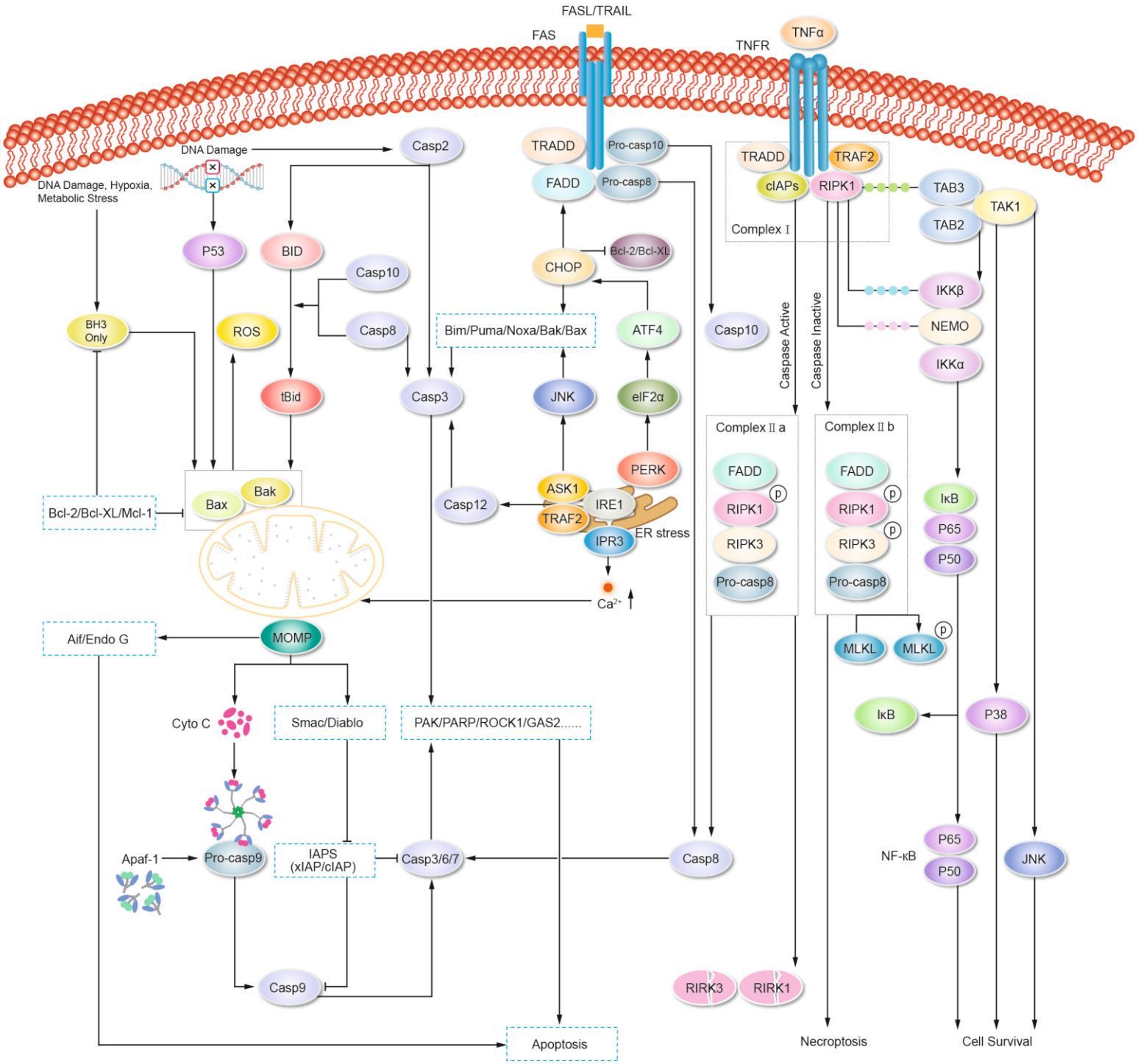
Pro-apoptotic		Anti-apoptotic	
Bax	Bid	Bcl-2	A1
Bok	Bik	Bcl-XL	Blfl-1
Bak	Noxa	Bcl-W	Bcl-B
Bim	Puma	Mcl-1	
Bad			

Caspase Superfamily

Initiator	Effector	Inflammatory
Caspase-2	Caspase-3	Caspase-1
Caspase-8	Caspase-6	Caspase-4
Caspase-9	Caspase-7	Caspase-5
Caspase-10		

Solarbio® LIFE SCIENCES

Apoptosis Signaling Pathway



图注 LEGEND

直接刺激	易　位	ub ubiquitination
直接抑制	多级刺激	p phosphorylation
间接刺激	结　合	间接抑制

Sri Saradi Scientific

Authorised Exclusive Partner of Solarbio in India.

Beijing Solarbio Science & Technology Co.,Ltd.

301, Third Floor, Building 26, No. 16, Huanke Middle Road, Beijing Economic-Technological Development Area, Beijing, China

info@solarbio.com

solarbio.com